

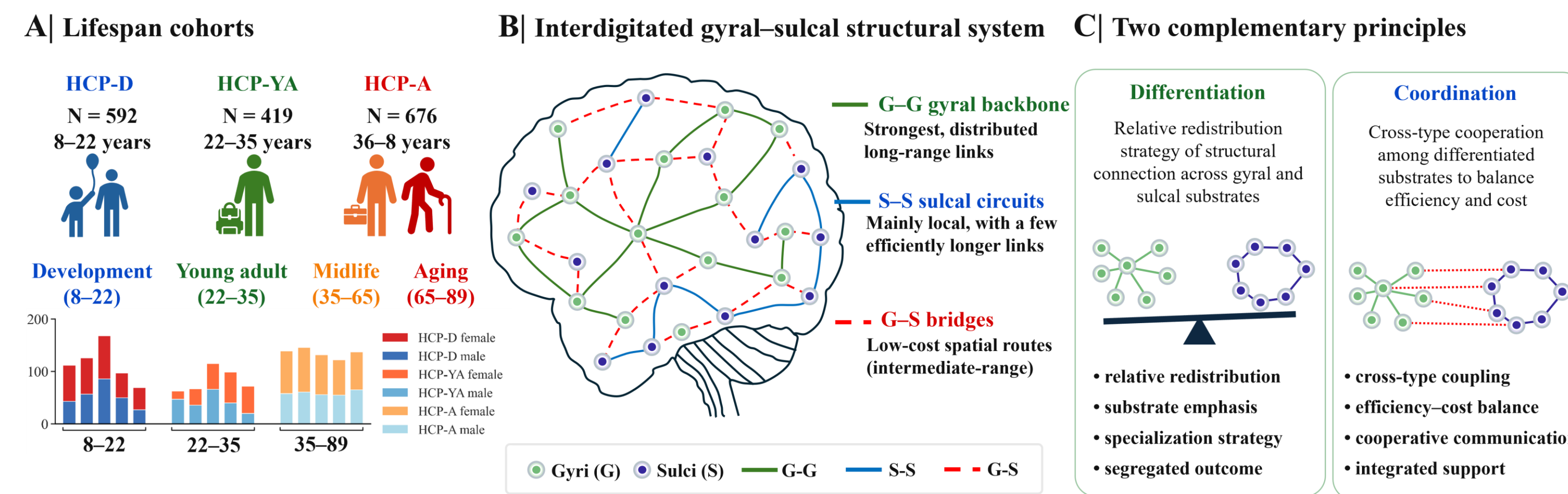
Differentiation and coordination of gyral–sulcal structural networks reveal lifespan cortical reconfiguration associated with executive function

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Background

Human brain networks undergo continuous and nonlinear reconfiguration. Most previous lifespan connectome studies have focused on absolute changes in integration and segregation, asking whether the brain becomes more globally integrated or more locally segregated with age. However, less is known about how structural resources are redistributed between anatomically distinct substrates (**differentiation**), and how these substrates remain coordinated across the lifespan (**coordination**). Gyri and sulci provide a folding-informed system for addressing this question.



Methods

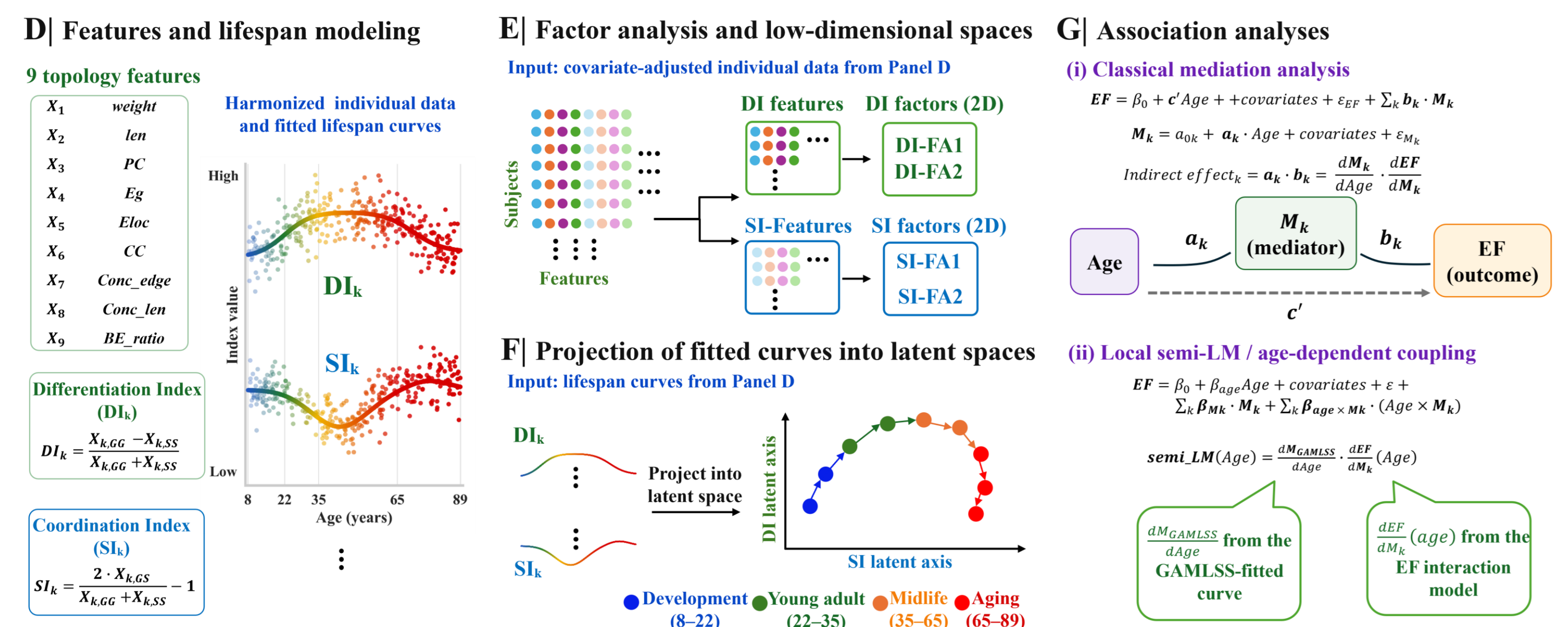


Figure 1. Overview of the analysis framework. **(A)** We analyzed structural connectomes across the HCP lifespan cohorts. **(B)** Structural connectomes were mapped onto an interdigitated gyral–sulcal system. **(C)** Differentiation quantified the relative redistribution of topological profiles between gyral and sulcal substrates, whereas coordination quantified cross-type gyral–sulcal coupling. **(D)** Features were defined, and nonlinear lifespan trajectories were modeled. **(E)** Covariate-adjusted individual features were embedded into low-dimensional latent spaces. **(F)** Fitted lifespan curves were then projected into these latent. **(G)** Classical mediation-style and local age-dependent coupling analyses were used to assess associations between latent structural trajectories and executive function

Results

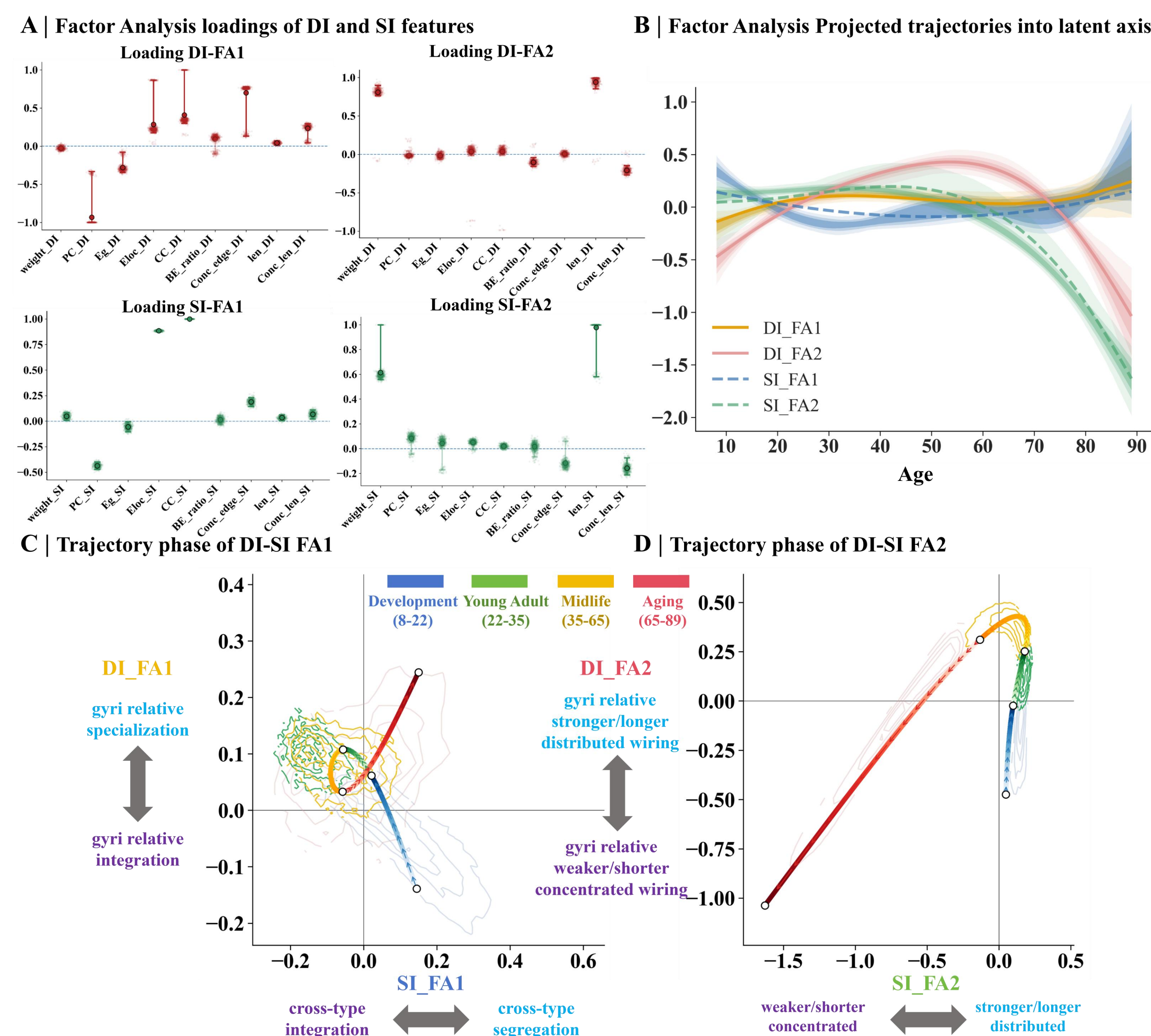


Figure 2. Latent axes and lifespan trajectories of DI/SI organization. **(A)** Factor analysis of covariate-adjusted individual features identified four latent axes: DI-FA1, DI-FA2, SI-FA1, and SI-FA2. **(B)** Fitted lifespan DI/SI curves were projected into the individual-defined latent spaces to obtain age-varying factor trajectories. **(C–D)** Phase-space plots summarize stage-dependent transitions in gyral–sulcal differentiation and coordination across development, young adulthood, midlife, and aging

Latent axes of gyral–sulcal differentiation and coordination

Index	FA1	FA2
Differentiation (DI)	relatively integration	wiring allocation, relative strength and length
Coordination (SI)	cross-type coordination	wiring scale of coordination

Cortical reconfiguration shows distinct stages across lifespan

Development was characterized by movement toward a relatively intermediate differentiation state and more efficient gyral–sulcal coordination. Young adulthood extended this trajectory with weaker fine-tuning. Midlife emerged as a transition window, suggesting rebalancing between gyral and sulcal substrates. Aging moved away from the intermediate differentiation balance and showed reduced cross-type coordination.

Wiring cost is conserved within age groups

Early-life EF were more aligned with differentiation-related wiring cost, whereas later-life EF were more closely associated with preserved cross-type wiring cost.

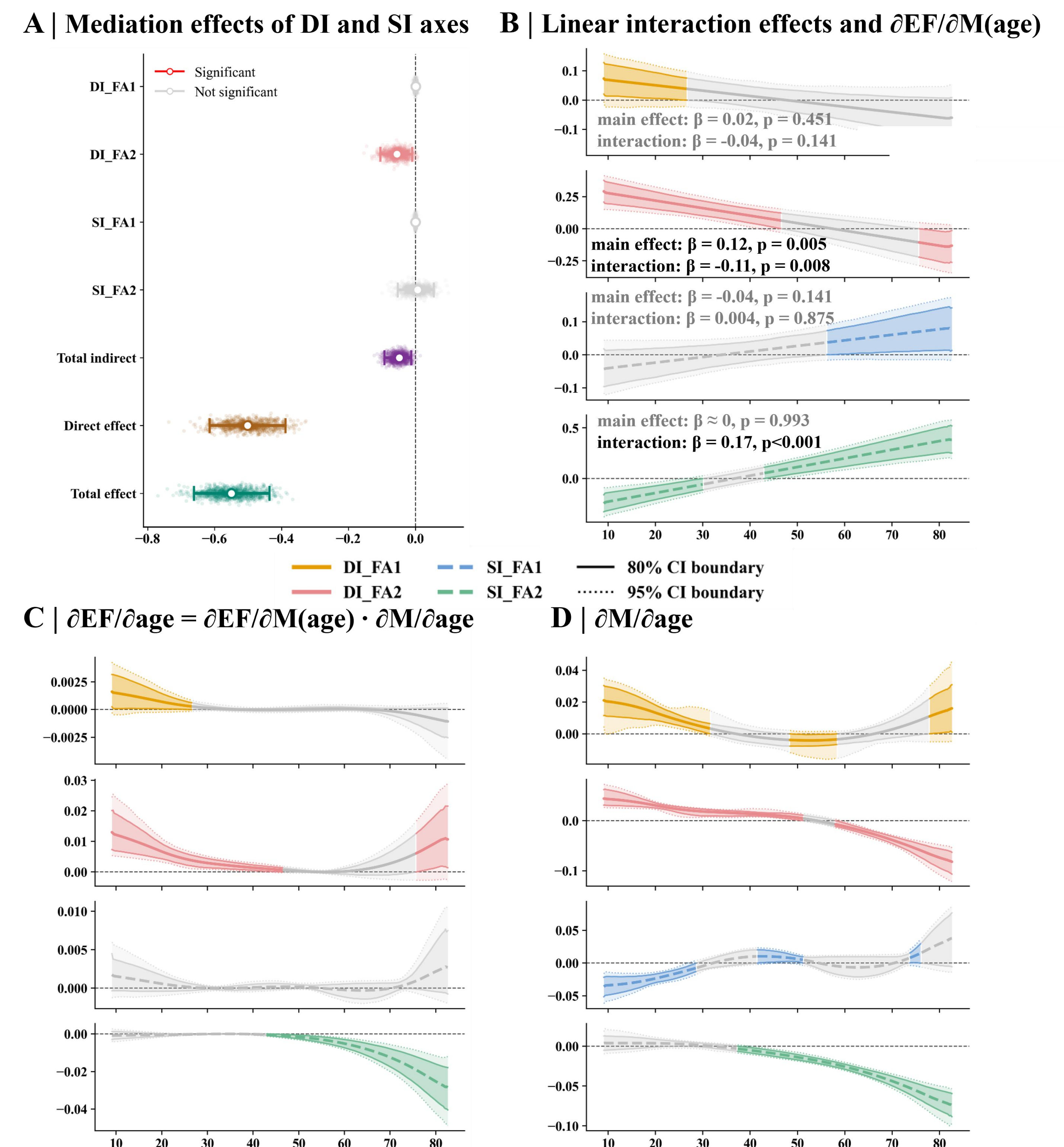


Figure 3. Linking latent gyral–sulcal reconfiguration to executive function. **(A)** Parallel mediation analysis assessed whether 4 latent axes accounted for part of the age–EF association. **(B)** Age-interaction models estimated the age-dependent association between each latent axis and EF. **(C)** A local coupling index combined age-related latent trajectory change with age-dependent EF coupling. **(D)** First derivatives of fitted latent trajectories were shown

Conclusion

This study provides a folding-informed characterization of lifespan cortical reconfiguration through gyral–sulcal differentiation and coordination. The identified latent trajectories reflect two canonical principles of brain network organization: **integration–segregation balance** and **cost-efficient wiring allocation**. Across the lifespan, these trajectories reveal distinct stages of cortical reconfiguration, including **developmental refinement**, **young-adult fine-tuning**, **midlife rebalancing**, and **aging-related reduction in cross-type coordination**. Together, these findings suggest that lifespan structural connectome reconfiguration reflects the dynamic redistribution and coordination of gyral and sulcal structural substrates.

References

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For lifespan or cortical folding! Looking for collaboration :)

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